and reconnaissance (ISR) missions and precision strike operations. Building upon the legacy of its predecessor, the RQ-1/MQ-1 Predator, the Reaper introduced significant improvements in speed, payload capacity, endurance, and lethality, establishing itself as a cornerstone of modern unmanned warfare (Cullen, 2011).

Design and technical overview

The MQ-9 Reaper, as shown in Figure 4.4, features a medium-altitude long-endurance (MALE) airframe made from lightweight composite materials, ensuring durability and aerodynamic efficiency. It has a wingspan of 20 meters (66 feet) and a fuselage length of 11 meters (36 feet). Powered by a Honeywell TPE331-10 turboprop engine producing 900 horsepower, the Reaper can reach speeds of up to 480 km/h (300 mph) and altitudes of 50,000 feet, nearly double that of the MQ-1 Predator. Its endurance of 27 hours allows it to conduct prolonged missions covering multiple operational objectives.



Figure 4.4 MQ-9 Reaper (2007).⏎